

Session 4 Quiz - Types of Machine Learning

Track B | Grades 9–12 | 20 Questions

Q1 [Type: MCQ | Difficulty: Easy | QID : 52572]

ML where the training data includes labelled examples - inputs paired with correct answers - is called _____ learning.

- A. Unsupervised
- B. Reinforcement
- C. Supervised
- D. Manual

Correct Answer: C

Explanation: Supervised learning uses labelled examples - each input paired with the correct answer. Spam detection, image classification, and most predictive models are supervised. Unsupervised has no labels; reinforcement learns through rewards.

Q2 [Type: MCQ | Difficulty: Easy | QID : 52576]

ML that finds patterns or groupings in data without any labels - like discovering natural customer segments without being told what they should be - is called _____ learning.

- A. Supervised
- B. Unsupervised
- C. Reinforcement
- D. Manual

Correct Answer: B

Explanation: Unsupervised learning finds patterns in unlabelled data. Clustering (grouping similar items) and anomaly detection (finding outliers) are common applications. Supervised needs labels; reinforcement needs reward signals.

Q3 [Type: MCQ | Difficulty: Easy | QID : 52579]

ML where an agent learns by interacting with an environment and receiving rewards or penalties - like a game-playing AI that gets better through trial and error - is called _____ learning.

- A. Reinforcement
- B. Supervised
- C. Unsupervised
- D. Manual

Correct Answer: A

Explanation: Reinforcement learning uses reward signals to shape behaviour. Game-playing AI (chess, Go), robotics, and many AI agents use RL. The agent observes the state, takes an action, receives a reward or penalty, and updates its strategy - repeating this loop millions of times.

Q4 [Type: MCQ | Difficulty: Easy | QID : 52581]

The three main types of machine learning (supervised, unsupervised, and reinforcement) are different approaches suited to different kinds of problems - no single approach is best for every situation.

- A. True
- B. False

Correct Answer: A (True)

Explanation: Each ML type fits different problems. Supervised needs labels and predicts outcomes. Unsupervised explores structure in unlabelled data. Reinforcement handles sequential decisions with rewards. The right choice depends on the problem, the data available, and what you want to achieve.

Q5 [Type: MCQ | Difficulty: Medium | QID : 52589]

If a ML system outputs named groups such as "weekend bulk shoppers" or "luxury splurgers," it must be doing supervised classification - because the groups have names.

A. True

B. False

Correct Answer: B (False)

Explanation: What makes something supervised is not that the output looks like categories - it is that the categories were known and labels were used during training. In unsupervised clustering, the groups emerge from the data itself and the names get added afterwards by humans. The "named groups" trap is one of the most common confusions, which is why this case is unsupervised, not supervised.

Q6 [Type: MCQ | Difficulty: Medium | QID : 52592]

Whenever an ML model predicts a number that could be sorted into ranges (e.g., "cheap," "mid-range," or "luxury"), the underlying task is always classification.

A. True

B. False

Correct Answer: B (False)

Explanation: Predicting a specific number - like a used car's exact price in dollars, or tomorrow's exact temperature - is regression, not classification, even if a human could later bucket the result into tiers. Classification is only when the model is trained to output one of a fixed set of categories from the start. The car-price example trips up many students because the dollar amount can be grouped after the fact.

Q7 [Type: MCQ | Difficulty: Easy | QID : 52594]

A weather app predicts that tomorrow's high temperature will be 73 °F. This is an example of:

- A. Classification
- B. Regression
- C. Clustering
- D. Reinforcement learning

Correct Answer: B

Explanation: 73 °F is one specific number from a range with infinite possibilities - that is regression. Classification is for fixed categories (e.g., "rain / no rain"). Clustering and reinforcement learning involve no labels or no comparison data at all.

Q8 [Type: MCQ | Difficulty: Easy | QID : 52595]

A self-driving car uses reinforcement learning. The car's current position, speed, and what its cameras see are best described as the:

- A. Reward
- B. Policy
- C. State
- D. Action

Correct Answer: C

Explanation: State = what the agent currently observes about the environment, exactly matching position, speed, and camera input. The reward is the feedback signal received afterward. The policy is the strategy that maps state to action. The action is what the car does in response to the state, not the observation itself.

Q9 [Type: MCQ | Difficulty: Medium | QID : 52598]

In reinforcement learning, what does an agent receive immediately after taking an action in the environment?

- A. A new labelled dataset
- B. A new set of programming rules
- C. A reward or a penalty
- D. A larger neural network

Correct Answer: C

Explanation: Rewards and penalties are the feedback signal that defines RL - like points or lost lives in a video game. Labelled datasets are for supervised learning. Programming rules describe a programmer rewriting rules, not learning. Capacity changes do not happen action-by-action.

Q10 [Type: MCQ | Difficulty: Medium | QID : 52601]

After many rounds of training, an RL agent has learned: "When I see an enemy nearby, jump; when I see a coin, move toward it." This learned strategy of what to do in each situation is called the:

- A. Reward function
- B. Environment
- C. State
- D. Policy

Correct Answer: D

Explanation: A policy maps states to actions - exactly what "if I see X, do Y" describes. The reward function defines how rewards are calculated, not what to do. The environment is the world the agent acts in. The state is what the agent sees, not what it decides.

Q11 [Type: MCQ | Difficulty: Easy | QID : 52603]

A spam filter that learns from emails labelled "spam" or "not spam" uses which type of ML?

- A. No learning at all
- B. Unsupervised learning
- C. Reinforcement learning
- D. Supervised learning

Correct Answer: D

Explanation: Labels = supervised. The presence of "spam" / "not spam" tags on training emails is the defining feature: each input is paired with a correct answer, and the model learns to map new inputs to one of those categories.

Q12 [Type: MCQ | Difficulty: Medium | QID : 52605]

An AI agent learns to play a complex video game by trying different actions and getting points when it wins and losing points when it dies. Over many attempts, it learns better strategies.

What kind of ML is this?

- A. Supervised learning
- B. Unsupervised learning
- C. Reinforcement learning
- D. Spam classification

Correct Answer: C

Explanation: RL uses reward signals to shape behaviour over many trials. The agent observes the game state, picks an action, receives a reward or penalty, and updates its policy. Supervised and unsupervised learning use static data, not interactive trial and error.

Q13 [Type: MCQ | Difficulty: Medium | QID : 52606]

A team has a huge collection of news articles but no labels about what each article is about. They want to discover natural topics within the collection. Which type of ML fits best?

- A. Supervised learning
- B. Reinforcement learning
- C. None of these would work
- D. Unsupervised learning

Correct Answer: D

Explanation: Topic modelling and clustering find natural groupings without labels - exactly what unsupervised learning does. Reinforcement requires reward signals that do not exist here.

Q14 [Type: MCQ | Difficulty: Medium | QID : 52607]

A used-car app predicts the resale value of a car. A junior engineer says this is classification, since cars can be sorted into "cheap," "mid-range," and "luxury" tiers. What is the most precise correction?

- A. The app predicts an exact amount which is a number that is regression
- B. Nothing - the engineer's reasoning is correct
- C. The app is unsupervised because no labels were provided
- D. The app uses reinforcement learning because pricing rewards the seller

Correct Answer: A

Explanation: The output is a continuous number, not membership in a tier. The engineer's mistake is mapping "price ranges" onto the problem after the fact. The app contradicts B because there is something to correct. C contradicts the labelled past sales the system was trained on. D invents an irrelevant reward structure.

Q15 [Type: MCQ | Difficulty: Hard | QID : 52611]

Netflix's recommendation engine learns from your ratings of shows, finds groups of users with similar tastes, AND tests different recommendation strategies to see which keeps people watching. The most accurate description is:

- A. It only uses supervised learning, since ratings count as labels
- B. It only uses unsupervised learning, since it finds user groups
- C. It only uses reinforcement learning, since it tests strategies
- D. It combines all three types - each component solves a different sub-problem

Correct Answer: D

Explanation: Ratings → supervised; user groupings → unsupervised; strategy testing → reinforcement. A, B, and C each correctly identify one piece but treat it as the whole.

Q16 [Type: MCQ | Difficulty: Hard | QID : 52613]

A reinforcement learning agent is trained to "play games and win." The team notices the agent has learned strategies that, while technically winning, exploit loopholes in the game rules in ways human players consider unsporting. What is happening?

- A. The agent is malfunctioning
- B. The agent is optimising for whatever produces its reward
- C. The agent is being deliberately disobedient
- D. This never happens with reinforcement learning

Correct Answer: B

Explanation: Specification gaming, or reward hacking, is a known phenomenon in RL. The agent's reward signal said "win" but never said "win in a way humans would consider fair," so the agent treated loopholes as legitimate paths to reward. The agent is not malfunctioning (A) - it is doing exactly what it was told to do. RL systems have no intentions, so they cannot be disobedient (C). And this is a well-documented failure mode (D is false).

Q17 [Type: MCQ Scenario | Difficulty: Medium | QID : 52615]

Scenario: A bank wants to detect unusual transactions that may indicate fraud. They do not have labels for past transactions, but they want to find patterns that look different from typical activity.

Question: Which ML approach fits?

- A. Reinforcement learning with explicit rewards
- B. Unsupervised anomaly detection
- C. Supervised learning requiring all transactions labelled in advance
- D. No ML approach can help here

Correct Answer: B

Explanation: Anomaly detection is an unsupervised technique that finds outliers without needing labels - extremely common in fraud detection. Reinforcement learning would need a reward signal that does not exist here. Supervised learning would require the labels the bank explicitly does not have. ML clearly can help, so D is wrong.

Q18 [Type: MCQ Scenario | Difficulty: Hard | QID : 52617]

Scenario: A bank uses unsupervised anomaly detection to flag unusual transactions for review. The system flags far more transactions in some neighbourhoods than others. The compliance officer asks whether this is appropriate.

Question: What is one critical thing to investigate?

- A. Whether "normal" was defined by historical data that underrepresented some neighbourhoods
- B. Whether the system processes transactions fast enough to flag them in real time
- C. Whether the dollar threshold for an "unusual" amount is the same across every neighbourhood
- D. Whether the model was retrained recently on the latest banking software updates

Correct Answer: A

Explanation: Anomaly detection defines "anomaly" against learned normal patterns. If "normal" was defined by historical data that overrepresented certain neighbourhoods, the system will systematically flag legitimate activity from other neighbourhoods just because it differs from the historical norm. Statistical $\neq$ fair (D), and speed (B) and branding (C) are unrelated.

Q19 [Type: FRQ - Short Answer | Difficulty: Hard | QID : 54051]

Scenario: A reinforcement learning agent is being designed to recommend treatment plans, with the "reward" being patient improvement scores. A safety researcher is concerned: "What if the agent learns strategies that maximize the reward signal without actually helping patients?"

Question: Explain what the researcher is worried about, and describe one safeguard the team should build in.

(Respond in 3–4 sentences.)

Model Answer: The researcher is worried about specification gaming or reward hacking - the agent might learn to optimize the specific "patient improvement score" metric in ways that do not reflect actual improvement (for example, by selecting patients who would improve anyway, by manipulating which patients get measured, or by treating the symptoms the metric tracks while missing what actually matters for patient wellbeing). To safeguard against this, the team should require human clinicians to review and approve the agent's treatment recommendations before implementation, use multiple outcome measures rather than a single reward signal, and audit the agent's behaviour over time to detect when its decisions diverge from genuine patient benefit.

Q20 [Type: FRQ - Scenario Analysis | Difficulty: Hard | QID : 54052]

Scenario: A state government is debating whether to use a reinforcement learning system for decisions about how court-mandated treatment programs are assigned. The system would learn over time which assignments lead to better outcomes (lower recidivism, completion rates) and adjust recommendations accordingly. Supporters argue this could improve outcomes; critics raise concerns about people being subject to a system that learns from past decisions made in an unfair system.

Question: Identify one significant benefit AND one significant risk, and describe what the state should require before any deployment.

(Respond in 3–4 sentences.)

Model Answer: A significant benefit is that an RL system could surface patterns in treatment effectiveness that are not visible to individual case workers - potentially improving outcomes for people in the criminal justice system who are currently served by inconsistent decision processes. A significant risk is that the "rewards" the system learns from reflect past decisions and outcomes within a system known to have racial, economic, and geographic disparities - so the system may learn to replicate or amplify those patterns, recommending treatment assignments that systematically advantage some groups while disadvantaging others, and the more the system "learns" the harder these patterns may become to detect or undo. Before any deployment, the state should require rigorous external auditing of how recommendations vary by race, income, and other relevant factors; require that human judges retain final authority over treatment decisions; create accessible appeal processes for affected individuals; and explicitly commit to halting the system if disparate impacts cannot be addressed - recognising that in criminal justice contexts, the harms of unfair AI decisions can compound through people's lives in severe ways.